

VII. Conclusion

1. Summary of Findings

1.1 Project Scope and Evidence Branches

The completed project established a connected research and implementation pipeline for shrimp-disease image analysis. Its first evidence branch addressed four-class, single-label classification in the fixed order Healthy, BG, WSSV, and WSSV_BG. Its second branch addressed disease-region instance segmentation with BG and WSSV instance masks, while Healthy images were retained as empty-label negative samples. These branches were supplemented by candidate-model screening, controlled robustness tests, qualitative explainability, leakage-aware evaluation, attention and Fourier investigations, and a native Android deployment path.

The evidence branches should not be collapsed into one performance claim. Classification measures image-level decisions, segmentation measures spatial localization and negative-image behavior, and the mobile study measures application-side runtime feasibility under a specific implementation. The final Android workflow is a conditional two-model cascade: the FP32 classifier always executes first, while the FP16 instance-segmentation model is invoked only for an above-threshold BG, WSSV, or WSSV_BG classification. Healthy and below-threshold outputs terminate after classification. The predicted class is therefore an application-control signal rather than verified field ground truth.

1.2 Classification Findings

Candidate screening covered 17 TIMM classifiers and 15 Ultralytics classification variants under the retained seed-42 partition. The completed TIMM ledger showed that ConvNeXt-Tiny-IN22K produced the strongest TIMM test Macro-F1 at 84.16% with 27.82M parameters, placing it outside the project-specific deployment preference of approximately 15M parameters or fewer. Within that preference, MobileNetV3-Large was the strongest retained TIMM row at 80.10% test Macro-F1 and 4.21M parameters. The Ultralytics screen subsequently selected YOLO26m-cls, which reached 89.02% Macro-F1 at 10.36M parameters in the retained representative screen. The timing ledgers are auxiliary screening evidence and are not interpreted as one cross-framework or Android latency benchmark.

The controlled classification reference was YOLO26m-cls trained with cross-entropy. The final selected configuration retained the same reference architecture and introduced the training-only ASL-LDAM objective together with SimAM-DCFR feature recalibration. Under the recorded fixed seed-42 protocol, accuracy increased from 89.02% to 91.33%, Macro-F1 increased from 89.02% to 91.01%, and Cohen's Kappa increased from 0.8505 to 0.881593 (85.05 to 88.16 on the displayed $\times 100$ scale). The Macro-F1 change was +1.99 percentage points. Because the comparison used one fixed seed, it is an observed improvement under that protocol rather than evidence of statistical significance.

The final result must also be interpreted with the retained artifact boundary. The primary prediction-level file, complete classwise report, and primary count and normalized confusion matrices were not retained for the 173-image test partition. Consequently, the overall improvement cannot be decomposed reliably into exact class transitions. Avoiding reconstruction from rounded aggregate values limits class-specific interpretation but preserves factual integrity.

Additional-source experiments further qualified the main finding. On EXT-3-Original, the pro-

posed configuration reached 91.29% Macro-F1 compared with 72.30% for CE. On SDI-4 + EXT-3-Original, however, CE reached 86.51% Macro-F1 while the proposed configuration reached 82.18%, a 4.33-point disadvantage for the proposed method. Both regimes retrained their models after source-wise splitting and partition-wise merging; they were not zero-shot inference by the original SDI-4 checkpoint. The opposite outcomes therefore indicate sensitivity to source composition, class priors, and acquisition differences rather than universal superiority.

Across the five retained aggregate classification corruptions, the proposed classifier exceeded CE in Macro-F1 and Kappa. The Macro-F1 gains were +7.32 points for impulse noise, +3.71 for Gaussian noise, +6.03 for contrast reduction, +5.66 for defocus blur, and +7.07 for low light. Absolute performance under impulse noise nevertheless remained limited, so these results indicate relative resilience within controlled synthetic stress tests rather than solved severe-noise behavior or guaranteed farm generalization. Grad-CAM++, HiResCAM, and EigenCAM remained qualitative activation inspections and are not disease masks, causal explanations, or clinical localization evidence.

1.3 Instance-Segmentation Findings

The segmentation branch confirmed that split design materially changes apparent localization performance. The convention-matched 1,149-image subset contains 416 filename-derived specimen groups and supports specimen-grouped allocation, but the expanded 1,452-image inventory adds 303 unmatched-name images whose filenames do not expose compatible specimen identity. Those added images were allocated by image-level stratified splitting. The original subset is therefore group-aware, whereas the expanded inventory must not be described as fully specimen-leakage-free.

Healthy-negative diagnostics added information that mask mAP alone could not provide. Healthy was not a positive segmentation mask class, and Healthy images contained no positive disease-region annotation. Measuring whether those images produced retained disease masks exposed false-positive behavior that positive-only localization metrics would hide.

The baseline and comparator screen showed why model selection was multi-objective. RTMDet-Ins-S reached 48.30% full-test mAP50, 54.90% labeled mAP50, and 18.70% labeled mAP50-95, but it produced a 100% Healthy false-positive rate and used 10.153M parameters. The retained YOLO11n-seg reference was substantially smaller at 2.877M parameters and provided much better Healthy-negative behavior. The final CA-to-SimAM refinement at P3/P4/P5 used 2,854,718 parameters and reached 49.71% full-test mAP50, 55.07% labeled mAP50, and 19.64% labeled mAP50-95. These results support YOLO11n-seg as the more appropriate lightweight foundation for this application, not a claim that it dominates every metric or every detector family.

The broader attention-family comparison remains configuration-level evidence rather than a strict one-factor ablation because augmentation, epoch budget, early stopping, initialization, software version, placement, and auxiliary objectives were not identical. The separate CA-to-SimAM placement study more narrowly supports the retained P3/P4/P5 insertion decision, but it does not remove the protocol differences among the other attention families.

Fourier evidence also remained condition-dependent. With no augmentation and the hidden hook disabled, W1 ($\sigma = 50$, $\alpha = 0.10$) increased labeled mAP50 from 23.54% to 34.51%, but under the recorded strong augmentation policy it reduced labeled mAP50 from 59.51% to 57.71%. The direct paired spectrum study used 24 held-out images and 12 matched stochastic draws per image, producing 288 paired global observations. W1 alone increased high-band energy share by +0.000670, whereas

the incremental W1 effect after strong augmentation was +0.000151, or approximately 22.5% of the original high-band effect. Edge-sensitive summaries retained different fractions of the W1 effect, including approximately 21.1% for Laplacian variance, 56.1% for mean gradient, and 79% for lesion-region gradient. The defensible interpretation is partial spectral redundancy under this seed-42 regime, not causal proof or multi-seed confirmation.

1.4 System and Deployment Findings

The final Android application integrated the complete conditional two-model bundle rather than a classifier-only prototype. The FP32 classification artifact accepts a [1, 224, 224, 3] input and produces four scores in the fixed order Healthy, BG, WSSV, and WSSV_BG. Disease-triggered branches invoke the FP16 YOLO11n-seg + CA-to-SimAM artifact for BG/WSSV instance masks. The same application build, the same two model artifacts, and the same configured four-thread CPU path were used across the three Android devices.

Each member recorded 21 application runs. On the vivo Y100, the classification-only/no-segmentation branch averaged approximately 5.016 s and the segmentation-triggered branch approximately 6.914 s. On the POCO M7 Pro 5G, the corresponding branch means were approximately 1.860 s and 2.641 s. The Samsung A32 log contained 19 segmentation-triggered runs averaging approximately 2.155 s and two below-threshold no-segmentation runs averaging 0.499 s; because it contained no Healthy branch and only two no-segmentation observations, that value must not be treated as a matched hardware comparison. Branch composition differed across devices, so overall device averages are observational workload summaries rather than a strict hardware ranking.

The operational archive retained 21 usable real-world shrimp photographs collected with the three team-owned Android cameras under varied lighting, background clutter, viewpoint, orientation, and blur/softness; one excessively dark image was excluded. These photographs show that the deployed pipeline was exercised on non-dataset inputs under practical capture variation. They do not provide field-accuracy evidence because no veterinary, laboratory, or other expert disease ground truth was assigned. The recorded timing is application-side operational elapsed time unless a narrower timer scope is proven. No iOS/iPhone device was tested.

1.5 Overall Interpretation

The overall contribution lies in the combination of controlled model selection, explicit validity boundaries, and engineering traceability rather than in an unbounded performance ranking. The classification branch now includes a complete TIMM screen, a retained Ultralytics comparison, an imbalance-aware training objective, feature recalibration, corruption testing, multi-source qualification, and qualitative XAI. The segmentation branch combines leakage-aware evaluation, Healthy-negative diagnostics, a lightweight multi-objective model-selection rationale, attention placement evidence, and a direct Fourier-spectrum interaction analysis. The deployment branch connects both final artifacts through a conditional Android cascade and documents three-device operational runtime.

Together, these elements support a traceable preliminary image-screening system. They do not establish clinical validity, universal robustness, controlled cross-device superiority, or farm-ready disease recognition.

2. Contributions and Reflections on the Project

2.1 Technical Contributions

The classification contribution combined three levels of evidence. First, a broad seed-42 candidate screen compared 17 TIMM classifiers with Ultralytics classification candidates under a common dataset partition, showing that predictive quality, parameter count, and speed could not be reduced to one ranking criterion. Second, YOLO26m-cls was retained as the controlled reference within the approximately 15M-parameter deployment preference. Third, the final model integrated the custom single-label softmax ASL-LDAM objective with SimAM-DCFR while preserving a four-logit inference contract. ASL-LDAM remained training-only, while the SimAM-DCFR texture and channel-gating paths introduced trainable feature-recalibration components.

For segmentation, the project added leakage-aware grouping, explicit Healthy empty-label negatives, a lightweight YOLO11n-seg foundation, a final CA-to-SimAM P3/P4/P5 refinement, RTMDet instance-segmentation comparators, and Fourier-domain diagnostics. The final segmentation model remained compact at approximately 2.855M parameters. The direct spectrum study further provided mechanism-oriented evidence explaining why W1 can become partly redundant once a strong augmentation policy already redistributes substantial high-frequency content.

2.2 Methodological Contributions

The direct classification comparison preserved one fixed split, class order, preprocessing representation, and evaluation scale. Macro-F1 was used as the primary outcome because the four-class dataset was imbalanced and accuracy could be dominated by larger classes. The report also disclosed that historical model-family screening inspected test-partition Macro-F1, preventing that ranking from being misrepresented as validation-only selection.

For segmentation, the methodology now distinguishes the fully filename-compatible grouped subset from the 303 unmatched-name additions, rather than overclaiming leakage control for the entire expanded inventory. Full-test and labeled-only localization, Healthy false-positive behavior, disease missed-image behavior, mask-count error, and HScore were reported separately so that positive-mask localization could not hide negative-image failures. RTMDet and Ultralytics recall values were also kept within their recorded evaluation definitions rather than treated as automatically interchangeable.

Artifact-aware reporting remained a core methodological principle. Missing primary prediction files, classwise reports, confusion matrices, exact final-classifier parameter count, and exact final-classifier serialized size were not reconstructed from neighboring values. The broader attention configurations were not relabeled as strict ablations, and the paired Fourier measurements were interpreted as diagnostics rather than causal proof.

2.3 Practical Contributions

The project carried the fixed four-class mapping into a deployable FP32 classifier and integrated it with the retained FP16 segmentation artifact in one Android control flow. The application records predicted class, confidence, ranked alternatives, mask output when generated, and runtime metadata while keeping local inference separate from persistence and Firebase synchronization.

Three-device testing and the three-camera operational image set strengthen the practical evidence beyond a single prototype screenshot. They demonstrate that the common application/model bundle executed across distinct Android hardware and naturally varied inputs. The evidence remains opera-

tional rather than diagnostic: no veterinary field labels were available, image identity/order was not standardized across devices, and iOS was not evaluated.

2.4 Project Reflection

Several practices materially improved the final thesis. Reconstructing and auditing the classification screen exposed the complete TIMM evidence instead of leaving candidate identities without retained metrics. Leakage analysis changed the interpretation of segmentation results by showing why convenient image-level splitting could be optimistic. Retaining Healthy negatives made false-positive behavior visible, while RTMDet comparison reinforced that a small gain in localization cannot be evaluated independently of negative-image failure and model complexity.

The project would have been stronger with a formal experiment registry from the beginning. Every run should have frozen its dataset manifest, environment, command, checkpoint hash, prediction file, evaluation report, and timing scope. Multi-seed runs, matched one-factor ablations, independent mask review, exact final-artifact audits, and stage-separated mobile timers would have reduced the remaining uncertainty.

3. Limitations and Future Work

3.1 Classification Limitations

The primary classification result uses one fixed seed and does not provide a run-to-run distribution or statistical significance. Candidate screening inspected test-partition Macro-F1, so model-family selection did not preserve a completely untouched selection-independent test. Although the TIMM screening ledger is now complete, its timing fields are treated as within-TIMM auxiliary metadata and are not assumed to share an identical timing harness with the retained Ultralytics measurements. The final primary prediction file, classwise report, and confusion matrices were not retained, which prevents exact error-transition analysis.

EXT-3-Original and SDI-4 + EXT-3-Original were retraining experiments rather than direct external inference. They therefore measure adaptation under changed source composition, not zero-shot generalization of the original checkpoint. The exact parameter count and serialized size of the final SimAM-DCFR classifier artifact also remain unverified and must not be borrowed from the plain YOLO26m-cls reference row.

3.2 Segmentation Limitations

Specimen identity is inferred from filenames. This supports group-aware allocation only for the 1,149 convention-matched images; the 303 unmatched-name additions cannot be assigned by compatible specimen identity, so the complete expanded split is not fully specimen-leakage-free. Masks were produced by one project annotator without independent expert review or inter-annotator agreement, and the deterministic WSSV-priority rule for overlapping project-authored polygons is an annotation convention rather than a biological statement.

The broader attention-family experiments use heterogeneous protocols and are not strict one-factor ablations. RTMDet-Ins-S provides a useful endpoint comparator, but its recall definition is not numerically identical to Ultralytics-exported recall, and its 100% Healthy false-positive rate prevents a simple mAP-only selection argument. No matched original U-Net experiment was completed. A standard U-Net performs semantic pixel classification rather than native instance segmentation, so a

fair future comparison requires a common manifest, common semantic metrics such as Dice/IoU, and an explicitly defined instance-extraction protocol if instance-level mAP is to be compared.

3.3 Robustness and Explainability Limitations

Classification and segmentation used different corruption families and reporting rules, so their robustness values cannot be pooled into one benchmark. Synthetic transformations do not reproduce every farm, camera, water condition, compression pipeline, or disease stage. The W1 spectrum study is based on the retained seed-42 sample and matched stochastic draws; it supports a mechanism-oriented interpretation of spectral overlap with augmentation but does not establish causality or multi-seed generality.

Grad-CAM++, HiResCAM, and EigenCAM remain qualitative. Heatmap agreement may reveal obvious body-focused behavior, but it does not establish causal reliance on a disease sign, and heatmaps are neither clinical evidence nor substitutes for independently reviewed segmentation masks.

3.4 Deployment Limitations

The revised deployment evidence covers three Android phones and the final conditional two-model pipeline, but it remains an observational runtime-feasibility study. The devices did not execute a standardized image-by-image workload, branch composition differed across logs, and the application timer may include decoding, preprocessing, initialization, postprocessing, mask reconstruction, and interface overhead. Overall device means therefore cannot be used as a strict hardware ranking or as isolated neural-network latency.

The real-world image set improves capture diversity but lacks veterinary or laboratory disease ground truth, so it cannot support mobile accuracy, sensitivity, specificity, disease prevalence, or clinical validity. No iOS/iPhone device was available. Formal farmer or veterinary usability testing, confidence-calibration evaluation, user comprehension of masks/confidence, thermal-stability testing, and systematic CPU/GPU/NNAPI/LiteRT delegate comparison also remain outstanding.

3.5 Prioritized Future Work

The next development cycle should prioritize the following evidence-preservation and validation steps:

1. Freeze immutable dataset inventories, split manifests, checkpoints, predictions, environment locks, conversion metadata, application versions, and cryptographic hashes for every reported run.
2. Use validation-only candidate selection, reserve an untouched final test partition, and run at least three to five independent seeds for the CE and proposed classification configurations.
3. Regenerate classwise reports, count and normalized confusion matrices, calibration analysis, and paired failure cases from frozen prediction files.
4. Audit the final SimAM-DCFR export directly to obtain its exact parameter count, serialized size, tensor metadata, and artifact hash instead of inheriting values from the plain YOLO26m-cls reference.
5. Conduct independent multi-annotator segmentation review and strengthen specimen identity with acquisition metadata and duplicate detection, especially for the unmatched-name subset.

6. Run strict matched attention and Fourier ablations under identical data, initialization, optimization, augmentation, software, checkpoint, and evaluation rules; repeat the direct spectrum analysis across multiple seeds.
7. Add a fair U-Net-derived comparator only under a protocol that reports common semantic metrics and a defined instance-extraction rule when instance-level evaluation is required.
8. Perform controlled same-image cross-device Android profiling with stage-separated timers, warm-up control, fixed thread count, device/app/model hashes, and supported CPU, GPU, NNAPI, and LiteRT delegate configurations.
9. Extend deployment to iOS only when a compatible device and reproducible runtime path are available; do not infer iOS behavior from Android evidence.
10. Conduct prospective expert-labeled field validation and a controlled usability study before any claim of operational disease-recognition effectiveness, and keep the system positioned as preliminary screening support until such evidence exists.

3.6 Closing Perspective

The revised evidence changes the final interpretation in three important ways. First, classification model selection is now supported by a complete TIMM screening ledger rather than by candidate names alone, while the final YOLO26m-cls choice remains bounded by the project's predictive-efficiency criteria and the approximately 15M-parameter deployment preference. Second, the segmentation conclusion is no longer based only on one YOLO family comparison: RTMDet provides an independent instance-segmentation endpoint, the CA-to-SimAM placement study supports the retained final refinement, and the Fourier spectrum analysis explains why conventional augmentation can reduce the incremental effect of W1.

Third, deployment evidence now concerns the actual classifier-plus-conditional-segmentation application on three Android devices. This is stronger operational evidence than a single classifier screen, but it also makes the remaining boundary clearer: branch-dependent application time, unmatched device workloads, and unlabeled field photographs cannot be converted into a universal latency or disease-accuracy claim. The most defensible next step is therefore not to broaden claims, but to improve evidence quality through frozen artifacts, controlled workloads, repeated seeds, independent annotation review, and prospective expert-labeled field validation.

The project establishes a traceable research and engineering foundation for lightweight shrimp-disease image screening. Its strongest result is not one isolated metric, but the combination of controlled classification evidence, leakage-aware segmentation, Healthy-negative evaluation, mechanism-oriented robustness analysis, and an operational two-model Android implementation with explicit limits on what the current evidence can establish.